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Cambridge IGCSE Combined and Co-ordinated Sciences

B5 Plant Nutrition

Condensed Version — Key Concepts and High-Probability Exam Questions

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Cambridge IGCSE Biology · The most important concepts and exam questions for your test

Must-Know Concepts

Photosynthesis and Chlorophyll

- **Photosynthesis** is the process by which plants synthesise carbohydrates from raw materials using energy from light.
- **Raw materials:** carbon dioxide (from the air) and water (from the soil). **Products:** glucose and oxygen.
- **Chlorophyll**, a green pigment in **chloroplasts**, **absorbs energy from sunlight** and **transfers** it to carbon dioxide and water, making them react to form glucose. It **transfers energy from light into chemical energy in carbohydrates**.
- Chloroplasts hold the chlorophyll (spread out on membranes so that a lot of sunlight reaches it) and the **enzymes** for photosynthesis.

How Plants Use Glucose

Use of glucose	Why
Respiration	Releases energy, e.g. for active transport of mineral ions into root hairs and for building proteins
Starch	Storage: insoluble, so it does not affect osmosis or take part in reactions; quickly turned back into glucose
Sucrose	Transport in the phloem to parts that cannot photosynthesise
Cellulose	Cell walls for new cells (straight chains of glucose; starch has spiral chains)
Nectar	Attracts pollinators (insects, bats or birds) that carry pollen between flowers
Amino acids, then proteins	Growth; needs nitrogen from nitrate ions
Chlorophyll	Absorbing light energy; needs nitrogen (from nitrate ions) and magnesium (from magnesium ions)

Nitrate and Magnesium Ions

Feature	Nitrate ions	Magnesium ions
Element supplied	Nitrogen	Magnesium
Why needed	To make amino acids, and then proteins (also chlorophyll)	To make chlorophyll
Deficiency	Weak growth, yellow leaves	Yellowing of leaves, often especially between the veins

- Both deficiencies cause **yellow leaves**, because both elements are needed to make chlorophyll, so the plant photosynthesises less and grows poorly.

Leaf Structure and Adaptations

Layer (top to bottom)	What it is like	Function
Cuticle	Waxy, transparent, waterproof	Reduces water loss; lets light through
Upper epidermis	Transparent, tightly packed cells, no chloroplasts	Lets light through; reduces water vapour loss
Palisade mesophyll	Tall, narrow cells packed with chloroplasts, near the top	**Main site of photosynthesis**
Spongy mesophyll	Rounded, loosely packed cells with chloroplasts and many **air spaces**	Carbon dioxide and oxygen diffuse through the air spaces
Vascular bundle (vein)	Xylem vessels and phloem tubes	Xylem brings water and mineral ions; phloem carries sucrose away
Lower epidermis	Usually no cuticle; contains the stomata	Rarely in sunlight, so it loses less water vapour
Stomata and guard cells	Openings, each between two guard cells (with chloroplasts)	Carbon dioxide in, oxygen and water vapour out; guard cells open and close the stoma

- A **large surface area** captures more sunlight and lets more carbon dioxide diffuse in (air is only **0.04%** carbon dioxide). Being **thin** lets light reach the inner cells and carbon dioxide diffuse to them quickly.
- **Routes:** light passes through the transparent cuticle and upper epidermis; carbon dioxide diffuses from the air → stomata → air spaces → palisade cells (never through the roots); water travels soil → roots → xylem in the veins, then into the leaf cells by **osmosis**.
- **Chloroplasts:** palisade cells have the most, spongy mesophyll cells many, guard cells a few; other epidermis cells and xylem have none.

Factors Affecting the Rate of Photosynthesis

Factor	Effect of increasing it	Reason
Light intensity	Rate rises until another factor limits it	More energy for the reactions
Carbon dioxide concentration	Rate rises until another factor limits it	More of a raw material
Temperature	Rate rises to an optimum, then falls quickly	Enzymes work faster, but are denatured at high temperatures
Quantity of chlorophyll	Faster rate	More light energy can be absorbed

- The **limiting factor** holds the rate back. On a graph the line **rises** while that factor is limiting, then **levels off** when another factor becomes limiting.
- Glasshouse growers control temperature, light and carbon dioxide (often raised from 0.04% to about 0.1%) so that none limits the rate, giving a **high yield**.

Key Terms

Term	Definition
Photosynthesis	The process by which plants synthesise carbohydrates from raw materials using energy from light

Term	Definition
Chlorophyll	A green pigment in chloroplasts that captures energy from sunlight and transfers it into energy in carbohydrates
Starch	An insoluble carbohydrate made by linking many glucose molecules into long, spiral chains; the plant's energy store
Sucrose	A sugar whose molecules are made of glucose and fructose linked together; the form in which plants transport carbohydrate
Cellulose	A carbohydrate made of glucose molecules linked in long, straight chains; used to build plant cell walls
Nitrate ions	Mineral ions absorbed from the soil through the root hairs; they supply the nitrogen needed to make amino acids, proteins and chlorophyll
Magnesium ions	Mineral ions absorbed from the soil; magnesium is needed to make chlorophyll
Palisade mesophyll	A layer of tall, narrow cells with very large numbers of chloroplasts near the top of the leaf; the main site of photosynthesis
Stoma (plural stomata)	An opening in the lower epidermis that lets carbon dioxide and oxygen diffuse in and out, and water vapour diffuse out
Variegated leaf	A leaf that has some white areas (no chlorophyll) and some green areas (with chlorophyll)
Destarching	Leaving a plant in the dark for long enough for it to use up its starch stores
Control	A standard sample that you use as a comparison, to find the effect of changing a variable
Limiting factor	The factor in shortest supply (light intensity, carbon dioxide concentration or temperature) that stops the rate of photosynthesis going any higher

Key Equations, Tests and Experiments

Photosynthesis Equations

- **Word equation:** carbon dioxide + water → glucose + oxygen ("sunlight" above the arrow, "chlorophyll" below it)
- **Balanced equation:** $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$ (6 carbon, 12 hydrogen and 18 oxygen atoms on each side)
- **Glucose molecules made** = number of CO_2 molecules ÷ 6, e.g. 12 ÷ 6 = 2.
- **Rate of photosynthesis** = quantity of oxygen produced ÷ time, e.g. bubbles per minute or cm^3 of gas per minute.
- **Mean** = (1st try + 2nd try + 3rd try) ÷ 3, e.g. (24 + 26 + 25) ÷ 3 = 25 bubbles per minute.

The Starch Test

Leaves turn some of their glucose into starch, so starch in a leaf shows that it has been photosynthesising.

1. Boil the leaf in water for about 30 seconds: this breaks down the cell membranes so iodine solution can reach the starch.
2. Turn off the flame: alcohol is flammable.
3. Put the leaf into alcohol in a boiling tube standing in the hot water: this removes the chlorophyll, so colour changes are easy to see.

4. Dip the brittle leaf into hot water to soften it.
5. Spread it on a white tile and add iodine solution: **blue-black = starch**; stays **orange-brown = no starch**.

Showing That Chlorophyll, Light and Carbon Dioxide Are Needed

Factor	How it is removed	Control	Test part after iodine
Chlorophyll	Variegated leaf: the white areas have none	The green areas	White areas stay orange-brown
Light	Black paper over part of a leaf, then a day in the light	The uncovered part	Covered part stays orange-brown
Carbon dioxide	Leaf sealed in a flask with potassium hydroxide solution (absorbs carbon dioxide)	Leaf in a flask of distilled water	Leaf stays orange-brown

- Each control turns blue-black, so all three factors are **necessary** for photosynthesis.
- For light and carbon dioxide the plant is **destarched** first (two days in a dark cupboard), so any starch found must have been made during the experiment.
- A control is the same as the test in every way except for the one factor being investigated.

Measuring the Rate with an Aquatic Plant

- Count the oxygen **bubbles** released from the cut stem in one minute or, more accurately, collect the gas and measure its volume.
- **Light intensity:** move the lamp to at least five distances (light intensity decreases as distance increases), with a transparent **heat shield** so that the lamp's heat does not change the temperature.
- Keep the temperature, carbon dioxide concentration, piece of plant, lamp and counting time the same; count three times at each distance and calculate a mean.
- **Result:** the closer the lamp, the more bubbles per minute, so the higher the light intensity, the faster the rate.
- **Temperature version:** water-baths (e.g. 10 to 50 °C); the rate rises to an optimum, then falls quickly.
- **Carbon dioxide version:** different concentrations of **sodium hydrogencarbonate** solution (e.g. 0 to 0.4%); the rate rises, then levels off.
- **Graph:** independent variable on the x-axis; axes labelled with units; even scales; neat crosses; ruled lines point to point.

High-Probability Exam Questions

Question 1

Describe how plants synthesise carbohydrates. (4 marks)

Answer:

- Plants make carbohydrates by **photosynthesis**, mainly in the palisade mesophyll cells of their leaves.
- Carbon dioxide diffuses into the leaf from the air through the stomata; water is absorbed from the soil by the roots and carried to the leaf in the xylem.
- In the chloroplasts, **chlorophyll** absorbs energy from sunlight.
- Chlorophyll transfers this energy to carbon dioxide and water, making them react; the reactions are catalysed by enzymes.

- The products are **glucose** (a carbohydrate) and **oxygen**, which is released.
- Equation: $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$

Any four of these points score the 4 marks.

Question 2

How many glucose molecules can be made from 12 molecules of carbon dioxide and 12 molecules of water? (2 marks)

Answer:

- The balanced equation, $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$, shows that 6 molecules of carbon dioxide and 6 molecules of water make 1 molecule of glucose.
- 12 ÷ 6 = **2 glucose molecules** (and 12 molecules of oxygen).

Question 3

Outline **two** subsequent uses of the carbohydrates made in photosynthesis. (4 marks)

Answer:

- Glucose is converted to **starch** for storage.
- Starch is insoluble, so it does not affect osmosis, and it can quickly be turned back into glucose when needed.
- Glucose is converted to **sucrose** for transport.
- Sucrose is carried in the phloem to parts that cannot photosynthesise, such as the roots, where it is changed back into glucose.

Other uses that score (use + reason): respiration to release energy; cellulose for cell walls; amino acids and proteins for growth (using nitrate ions); nectar to attract pollinators.

Question 4

A drawing of a section through a leaf is labelled from top to bottom: A, the thin covering on the top surface; B, the top layer of cells; C, tall, narrow cells packed side by side; D, rounded cells with spaces between them; E, a bundle of tubes; F, a space between the rounded cells; G, a pore in the bottom surface; H, one of the two cells around that pore; I, the bottom layer of cells.

(a) Identify the tissue labelled C, and outline how this tissue is adapted for its function. [3]

(b) Give the letters of two cells or tissues that do **not** contain chloroplasts. [2]

(c) Describe the function of the cell labelled H. [2]

(d) Use the letters on the diagram to describe how water reaches a chloroplast in the leaf. [2]

(9 marks)

Answer:

- (a) C is the **palisade mesophyll**.
- (a) Its cells contain very large numbers of chloroplasts, to absorb as much light energy as possible.
- (a) It is just below the transparent upper epidermis, so it receives plenty of sunlight. Also accepted: tall, narrow, tightly packed cells; close to air spaces and veins.
- (b) **B**, the upper epidermis.
- (b) **I**, the lower epidermis (apart from its guard cells), or **E**, the vascular bundle. A, F and G are not cells.
- (c) H is a guard cell: it changes shape to open and close the stoma, G.
- (c) This controls the diffusion of carbon dioxide in and oxygen out, and the loss of water vapour.
- (d) Water arrives in the xylem vessels inside the vascular bundle, **E**.
- (d) It moves by osmosis through the spongy mesophyll, **D**, to the palisade cells, **C**, and into the chloroplasts.

Question 5

In the starch test, a leaf is boiled in water and then put into alcohol before iodine solution is added.

(a) Cell membranes are partially permeable: they will not allow starch or iodine molecules to pass through them. Use this information to explain why it is necessary to boil the leaf. [2]

(b) Suggest why it is a good idea to remove the chlorophyll before adding iodine solution. [2]

(4 marks)

Answer:

- (a) Iodine cannot pass through the intact cell membranes, so it could not reach the starch inside the cells.
- (a) Boiling breaks down the cell membranes, so the iodine solution can get into the cells and react with the starch.
- (b) The dark green colour of chlorophyll would hide (mask) the colour of the iodine solution.
- (b) Removing it leaves the leaf pale, so the blue-black colour change is easy to see.

Question 6

To show that light is needed for photosynthesis, a potted plant was destarched (kept in a dark cupboard for two days). Part of one leaf was then covered with black paper and the plant was left in the light for a day before the leaf was tested for starch. Suggest why destarching was a necessary part of this experiment. (3 marks)

Answer:

- Destarching makes sure there is no starch in the leaf at the start.
- So any starch found at the end must have been made by photosynthesis during the experiment, in the parts that received light.
- If the leaf still contained old starch, the covered part might test positive even though it had not photosynthesised, so you could not tell whether light is needed.

Question 7

To show that carbon dioxide is needed, two leaves on a destarched plant were sealed, still attached, in two flasks: one with potassium hydroxide solution (which absorbs carbon dioxide), the other with distilled water. After at least a day in the light, both leaves were tested for starch.

(a) What was the control in this experiment? [1]

(b) Suggest why distilled water was added to one flask, rather than just leaving it empty. [2]

(3 marks)

Answer:

- (a) The leaf in the flask of distilled water: it had carbon dioxide around it, but was otherwise treated in exactly the same way.
- (b) So that the two flasks were the same in every way except for the presence of carbon dioxide.
- (b) Potassium hydroxide solution is mostly water and makes the air humid; distilled water keeps the humidity the same, so any difference between the leaves can only be caused by carbon dioxide.

Question 8

An aquatic plant in a beaker of water is lit by a lamp at different distances, and the oxygen bubbles it releases in one minute are counted. Lamps produce heat as well as light. Explain why you would be unable to draw a conclusion about light intensity if you did not use a heat shield. (3 marks)

Answer:

- Without a heat shield, the lamp would warm the water more when it was close to the beaker than when it was far away.

- Temperature also affects the rate of photosynthesis, because it changes the activity of enzymes.
- So light intensity and temperature would both change at the same time, and you could not tell whether a change in the number of bubbles was caused by the light or by the heat.

Question 9

Two students put a piece of aquatic plant into each of five syringes of sodium hydrogencarbonate solution (which provides carbon dioxide) at different concentrations, and placed them in bright sunshine. Oxygen released by each plant collected in its syringe and pushed liquid down a transparent tube attached to the nozzle. The students recorded the distance moved by the meniscus in each tube after 30 minutes.

Percentage concentration of sodium hydrogencarbonate solution	0	0.1	0.2	0.3	0.4
Distance moved by meniscus after 30 minutes / mm	5	8	12	14	17

(a) Identify the independent variable and the dependent variable. [2]

(b) Explain why the five syringes were placed in bright sunshine. [2]

(c) Draw a line graph of these results, joining the points with ruled straight lines. [5]

(d) Explain the results of this investigation. [2]

(e) Suggest two possible sources of error, and how the investigation could be modified to reduce them. [4]

(15 marks)

Answer:

- (a) Independent variable: the concentration of sodium hydrogencarbonate solution (how much carbon dioxide is available).
- (a) Dependent variable: the distance moved by the meniscus in 30 minutes, which shows how much oxygen was produced, so the rate of photosynthesis.
- (b) Light provides the energy for photosynthesis.
- (b) Bright sunshine gives a high light intensity, so light is not limiting and the rate is limited by carbon dioxide; every syringe gets the same light, so it is a fair test.
- (c) Concentration on the x-axis and distance on the y-axis, both labelled with units (% and mm).
- (c) Even scales using at least half the grid, e.g. 0 to 0.4 in steps of 0.1, and 0 to 18 mm in steps of 2 mm.
- (c) All five points plotted accurately as small, neat crosses: worth 2 marks, or 1 mark if three or four are correct.
- (c) Ruled straight lines from cross to cross, not extended to the origin.
- (d) As the concentration increased from 0% to 0.4%, the distance moved increased from 5 mm to 17 mm.
- (d) Sodium hydrogencarbonate provides carbon dioxide, a raw material and the limiting factor here, so the rate of photosynthesis increased and more oxygen was produced.
- (e) Error: temperature was not controlled; syringes in sunshine may warm up, so the gas expands and the rate changes.
- (e) Improvement: stand the syringes in a thermostatically controlled water-bath.
- (e) Error: only one syringe for each concentration, so there were no repeats.
- (e) Improvement: set up three syringes for each concentration and calculate a mean.

Common Mistakes to Avoid

- Writing light or chlorophyll as reactants after a "+". They go on the arrow.

- Writing that plants or chlorophyll "make energy". Chlorophyll absorbs light energy and transfers it into chemical energy in glucose.
- Saying photosynthesis "changes carbon dioxide into oxygen". Carbon dioxide **and water** make glucose and oxygen; the carbon goes into the glucose.
- Saying carbon dioxide enters through the roots, or that plants get their food from the soil. Carbon dioxide diffuses in through the stomata; the soil supplies only water and mineral ions.
- Mixing up the ions: nitrate supplies nitrogen for amino acids and proteins; magnesium is needed for chlorophyll.
- Saying plants transport glucose. They change it into sucrose, which is carried in the phloem.
- Forgetting the control, the destarching or the heat shield in experiment answers, or treating bubble counts as exact (bubbles vary in size; measuring gas volume is more accurate).
- Losing graph marks with missing units, uneven scales, or a line extended beyond the data.

Last-Minute Checklist

- I can write the word and balanced equations for photosynthesis, with light and chlorophyll on the arrow.
- I can explain how chlorophyll transfers energy from light into chemical energy in glucose.
- I can give each use of glucose with a reason, including why starch is a good store.
- I can state why plants need nitrate and magnesium ions, and the deficiency symptoms.
- I can name the leaf layers from top to bottom, give their functions, and say which cells contain chloroplasts.
- I can explain how a palisade cell is adapted, and how carbon dioxide, water and light reach it.
- I can describe the starch test and give the reason for each step.
- I can describe how to show that chlorophyll, light and carbon dioxide are needed, naming each control and explaining destarching.
- I can describe how to investigate the effect of light intensity or temperature on the rate of photosynthesis, including the heat shield and repeats.
- I can draw a line graph that earns every mark, then describe and explain the trend using the data.